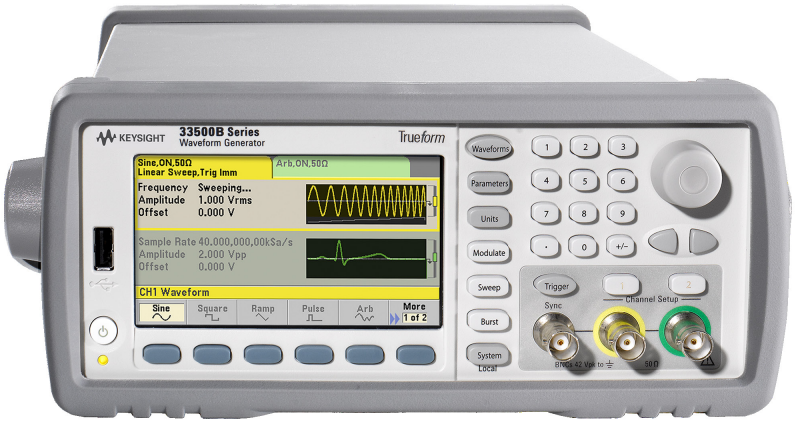
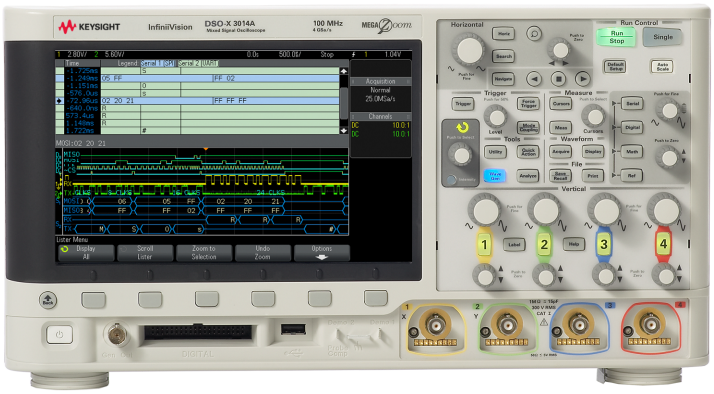
HANDS-ON LABORATORY / 60 MINUTES

One cable. Four discoveries.

Amplitude, frequency, pulse width and triggered bursts - foundations for quantum control.



Keysight 33500B Series



Keysight DSO-X 3014A

Begin with a visible sine wave. Measure its voltage and timing. Make rectangular pulses, then capture a short burst of sine-wave cycles. Use the measured burst duration in an explicitly hypothetical qubit model.

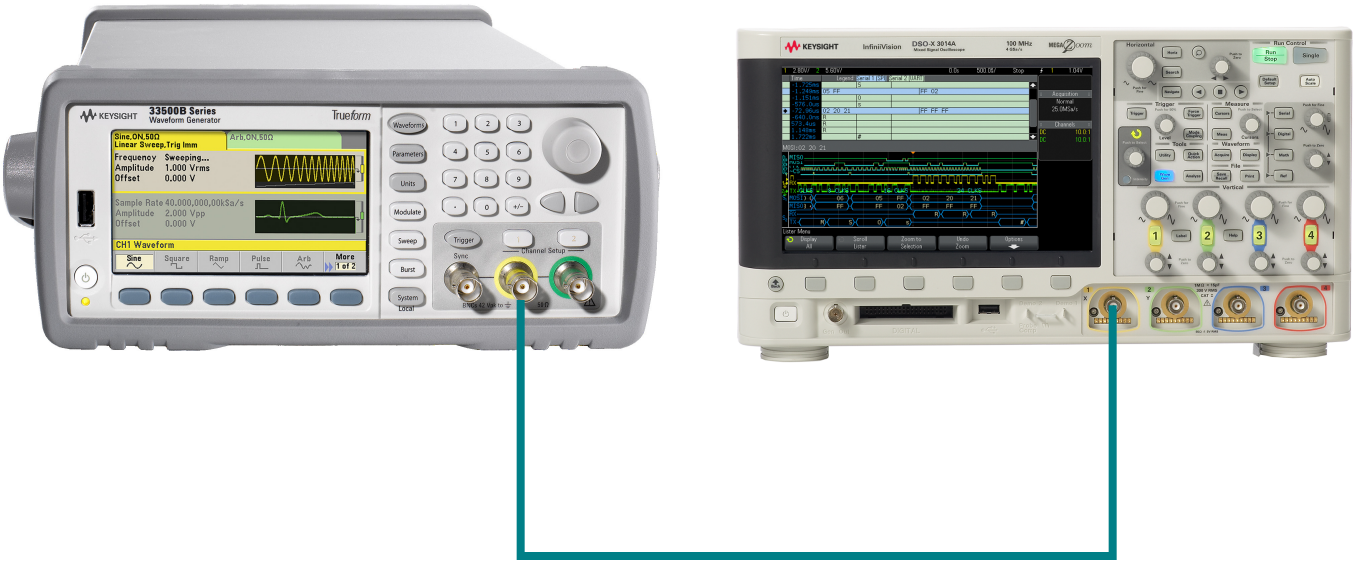
MEASURED SIGNALS + A SEPARATE QUANTUM MODEL

The instruments measure classical voltage versus time. They do not create a qubit. The quantum connection is how calibrated pulse strength, frequency and duration can control a real quantum system. Model predictions are labeled throughout.

01 / CONNECTIONS

One BNC from CH1 to CH1.

Use one generator, one oscilloscope and one BNC-to-BNC cable. Leave generator CH2 OFF.



Generator CH1 OUT → oscilloscope analog INPUT 1

1 Connect while generator Output is OFF

Locate the yellow-ring CH1 output on the generator. Push the BNC connector onto it and twist the collar to lock. Attach the other end to scope analog input 1.

2 Keep the other connectors unused

Do not use generator Sync, CH2, scope Gen Out or an external trigger input. The scope will trigger from the signal carried by this same cable.

BEFORE YOU START

Keep protective earth intact. Connect only these two instruments; no other powered circuit is needed. Official product photos identify the panels; screen contents in those photos are not the lab settings.

02 / READ VOLTAGE AND TIME

The grid is your ruler.

Your Keysight displays the voltage and time scale readouts at the TOP of its screen.

Quantity	Meaning	Units
Vpp	Maximum voltage minus minimum voltage. A centered 2 Vpp sine reaches +1 V and -1 V.	Volts (V)
Frequency f	Complete cycles per second. 1 kHz = 1,000 Hz.	Hz or kHz
Period T	Time for one complete repetition; measure between consecutive matching points.	s, ms or µs
Pulse width τ	Time a rectangular pulse stays high; different from the full repetition period.	µs or ms

FREQUENCY ↔ PERIOD

T (s) = 1 / f (Hz) | T (ms) = 1 / f (kHz)
T (µs) = 1,000 / f (kHz)
1 second = 1,000 ms = 1,000,000 µs.

Frequency	One-cycle period	At 200 µs/div
500 Hz	2 ms	10 large squares
1 kHz	1 ms	5 large squares
2 kHz	500 µs	2.5 large squares

03 / FIRST STABLE SINE WAVE

Set up once, then measure.

Buttons below the display select the softkey labels directly above them.

1 Select generator CH1 and its output load

Press the Channel Setup key 1 above the yellow BNC output, NOT the keypad digit. Keep Output OFF. Find Output Load / Load; use More if needed. Select High Z FIRST. [1]

2 Enter a continuous sine wave

Press Waveforms → Sine, then Parameters. Set Frequency 1.000 kHz, Amplitude 2.00 Vpp, Offset 0 V and Phase 0°. Leave Modulate, Sweep and Burst OFF. In CH1 Channel Setup set Output ON.

3 Configure the scope input

Enable input 1. Set 1 MΩ impedance, DC coupling and probe factor 1:1 for the direct BNC. Ensure Run is active. Use AutoScale to find the wave, then set 500 mV/div and 200 µs/div.

4 Stabilize the image and add measurements

Use Edge trigger, Source 1, Rising slope, Level 0 V. Press Meas; set Source 1 and Type Frequency, then Add Measurement. Add Period and Peak-to-Peak in the same way. [2]

PASS CHECK

Expect approximately 1 kHz, 1 ms and 2 Vpp.
At 500 mV/div: 2 Vpp spans 4 vertical squares.
At 200 µs/div: one cycle spans 5 horizontal squares.

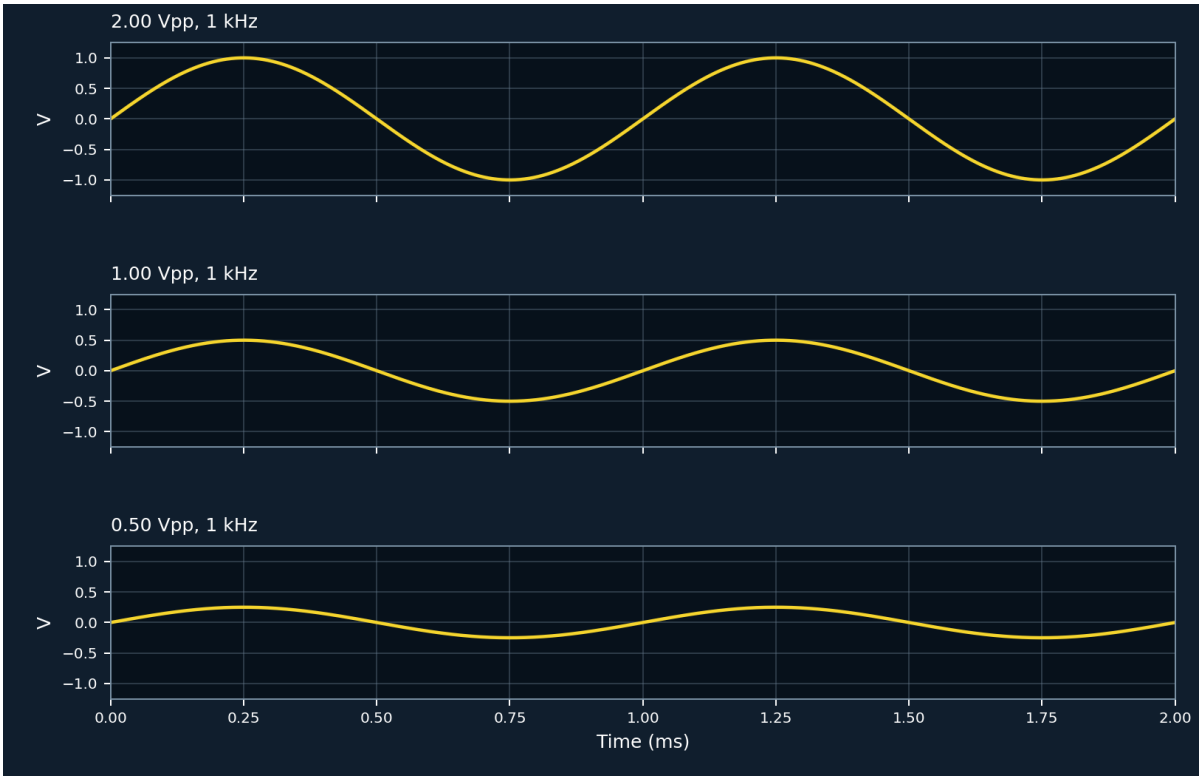
“Output stays ON” means leave the physical output enabled. It is an instruction, not another menu item. Menu wording may differ slightly by firmware.

04 / EXPERIMENT A

Amplitude: change the height

Keep frequency = 1 kHz, offset = 0 V and scope vertical scale = 500 mV/div.

Do: Parameters → Amplitude. Enter 2.00, 1.00 and 0.50 Vpp in turn. Record measured Peak-to-Peak. Do not use AutoScale between trials.



Set amplitude	Expected height	What stays fixed
2.00 / 1.00 / 0.50 Vpp	4 / 2 / 1 vertical squares	1 kHz frequency; 1 ms period

QUANTUM CONTROL CONNECTION

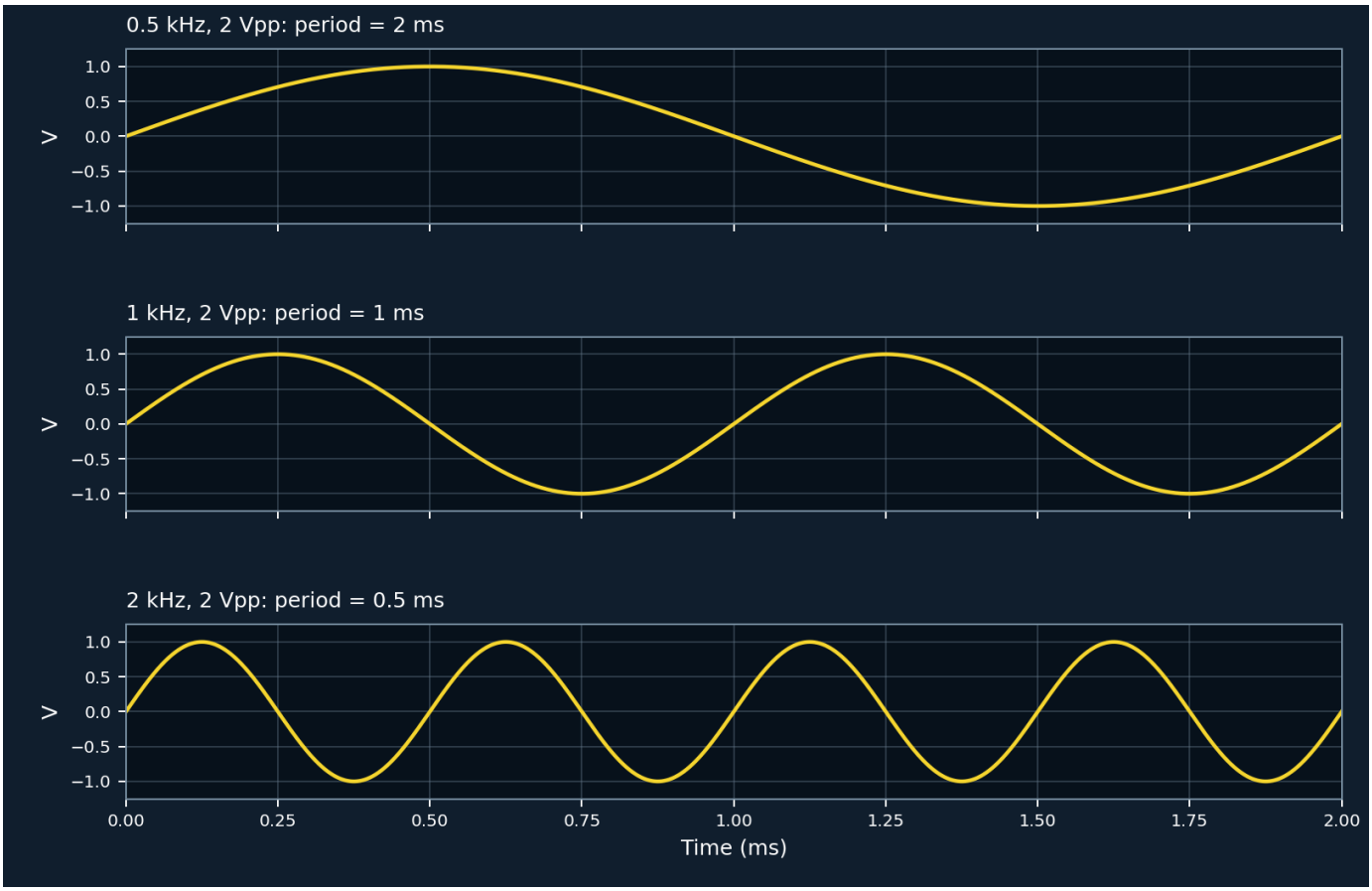
Drive strength can change a real qubit rotation under a calibrated model. Halving generator voltage alone does NOT establish a 25% quantum probability. Here we measure only electrical amplitude. [3]

05 / EXPERIMENT B

Frequency: change the spacing

Set Sine, 2.00 Vpp and 0 V offset. Keep the horizontal scale at 200 µs/div.

Do: Parameters → Frequency. Try 500 Hz, 1 kHz and 2 kHz. Read Frequency and Period on the scope. Higher frequency means a shorter period.



Check at 2 kHz: 1 / 2,000 = 0.0005 s = 500 µs.
Measure 2.5 horizontal divisions × 200 µs/div = 500 µs.
Voltage height stays approximately constant.

QUANTUM CONTROL CONNECTION

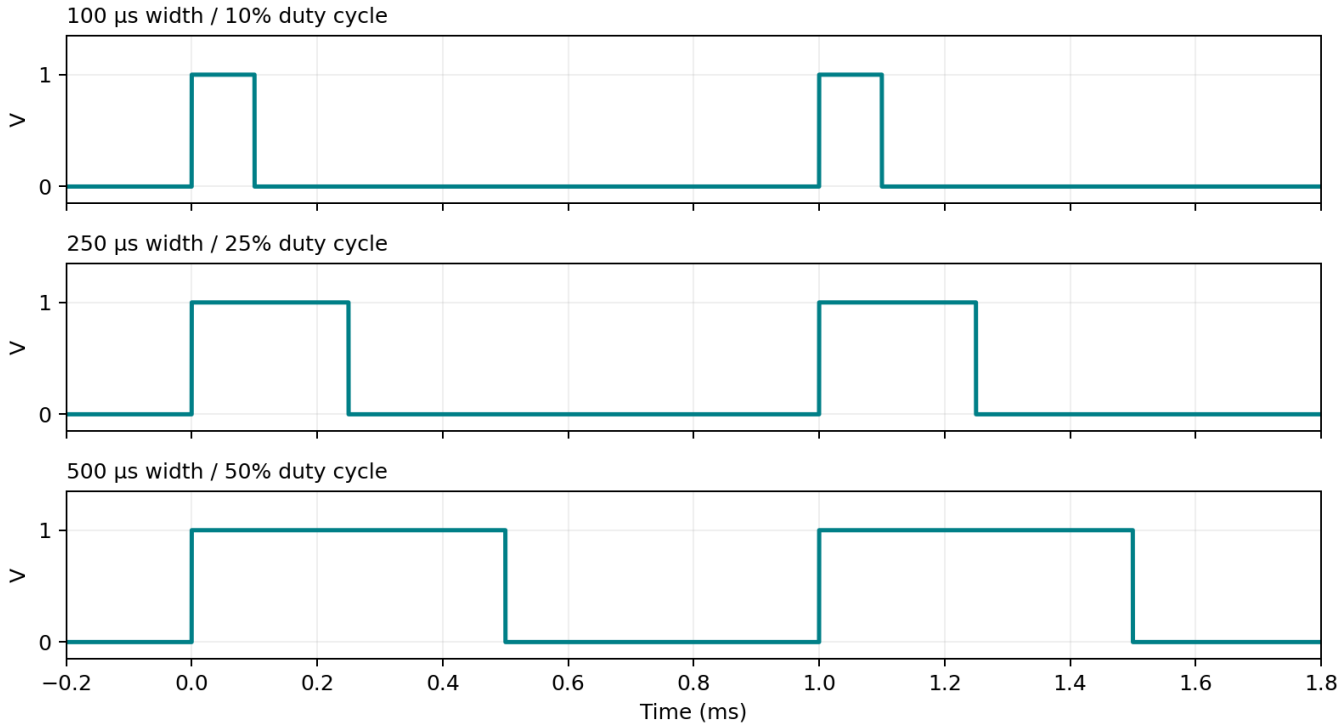
A real qubit drive must be appropriately tuned to the system. This kilohertz wave demonstrates timing, not an observed quantum transition or resonance. Doubling frequency does not itself demonstrate a quantum effect. [3]

06 / EXPERIMENT C

Pulse width: change time high

A pulse is a high interval followed by a low interval. Frequency sets the full repetition period.

Generator: Waveforms → Pulse; frequency 1 kHz; amplitude 1 Vpp; offset +0.5 V. Parameters → Width: 100, 250 and 500 µs. Use valid short edge times compared with 100 µs. Burst OFF; Output ON.
Scope: 200 mV/div, 200 µs/div; CH1 rising trigger at +0.5 V. Add positive pulse width measurement.



Expected: low ≈ 0 V, high ≈ +1 V. At 200 µs/div, the high widths span 0.5, 1.25 and 2.5 squares. The full period remains 1 ms.
Duty cycle: width / period × 100% = 10%, 25%, 50%.

QUANTUM CONTROL CONNECTION

Control-pulse duration matters in real qubit hardware. These rectangular pulses demonstrate timing. Their duty cycle is a classical time fraction, NOT a quantum measurement probability. [3]

07 / EXPERIMENT D: TRIGGERED BURST

Send a finite packet of cycles.

A burst contains N complete sine cycles. One cable carries both the signal and its trigger edge.

1 Change back to Sine, then enable Burst

Select Sine, 1.000 kHz, 1.000 Vpp, offset 0 V. Press Burst; choose triggered N Cycle operation (not Gated). Set # Cycles = 1 and burst start phase = 0°. Enable Burst. [1]

2 Choose Manual trigger on the generator

Open the trigger setup / source menu for the burst and choose Manual rather than Immediate or Timer. Keep CH1 Output ON and CH2 OFF. Each generator Trigger press should now send one burst. [1]

3 Prepare a single capture on the scope

Set 200 mV/div and 1 ms/div. Edge trigger: CH1, rising, level +0.10 V. Place the trigger marker about 10-20% from the left edge to leave room for the burst.

4 Arm FIRST, then send

Press Single on the OSCILLOSCOPE. Then press Trigger on the GENERATOR. The scope should capture and freeze one sine cycle near ±0.5 V, with a quiet baseline before and after.

REPEAT THIS ORDER FOR EVERY TRIAL

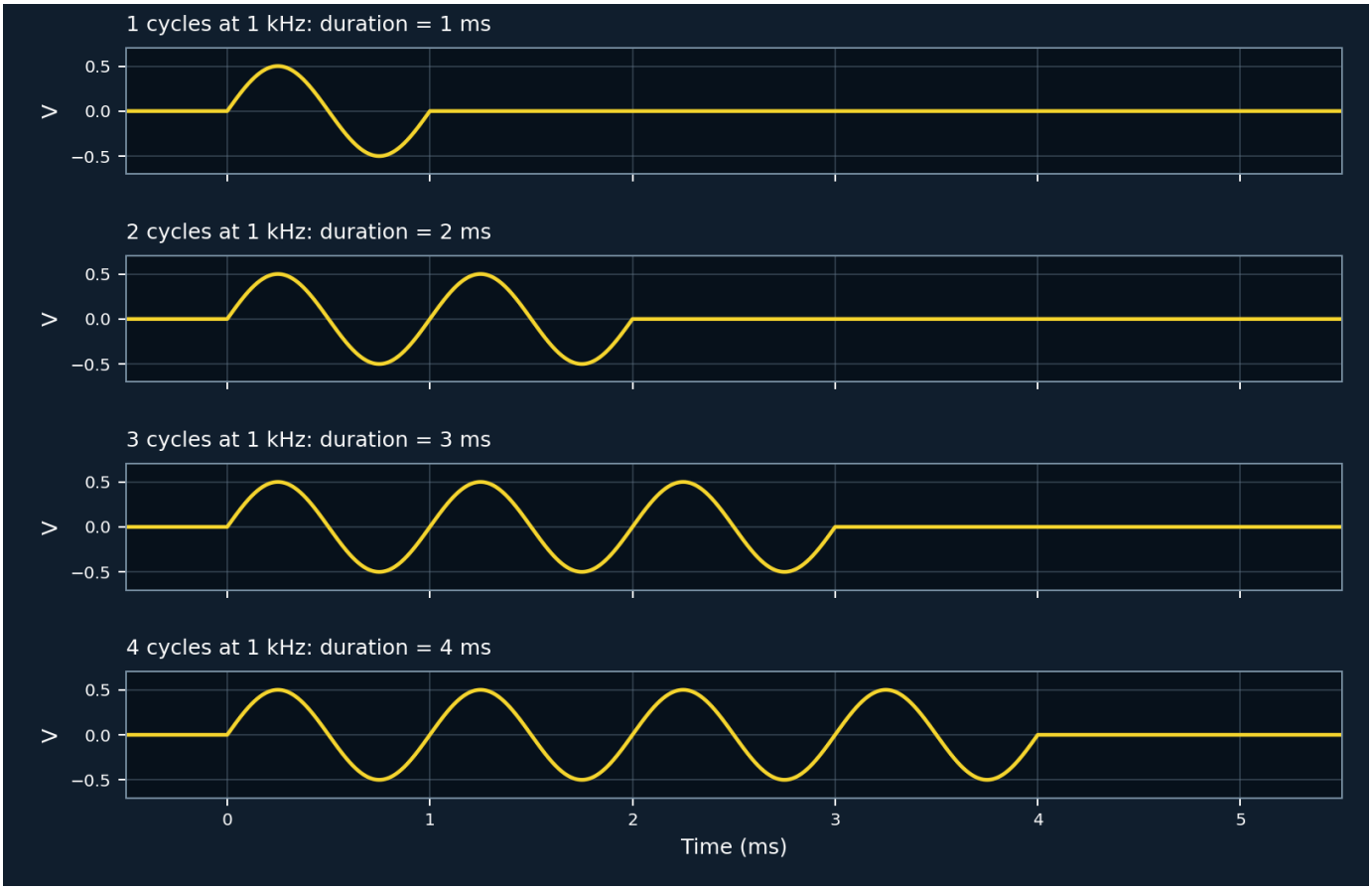
Set generator # Cycles → press scope Single → press generator Trigger → measure.
Try N = 1, 2, 3 and 4. Keep carrier frequency, amplitude and start phase fixed. The frequency stays 1 kHz while the packet becomes longer.

The scope trigger level is +0.10 V, so the start of the physical burst is slightly before the trigger crossing. Include that short pretrigger portion when measuring the complete packet.

08 / EXPECTED BURST WAVEFORMS

More cycles, longer duration

These are ideal calculated bursts, not captures from the instrument.



Measure: use time cursors from the start of the FIRST cycle to the end of the LAST complete cycle. First positive peak to last positive peak misses one full cycle.
Formula: burst duration $\tau = N / f$. At 1 kHz, N cycles last N milliseconds.

Number of cycles	1	2	3	4
Expected duration	1 ms	2 ms	3 ms	4 ms

09 / QUANTUM MODEL: EXPLICITLY ASSUMED

From duration to a rotation

A real resonant drive can rotate a qubit. Our bench does not measure that rotation. [3, 4]

CLASSROOM ASSUMPTIONS - NOT AN INSTRUMENT CALIBRATION

Assume an initially prepared $|0\rangle$ qubit, an ideal resonant X-axis drive and no noise. Choose a hypothetical Rabi rate $\Omega/(2\pi) = 250$ Hz for the fixed 1 Vpp drive. This rate is invented for teaching, not measured by the scope.

Model: rotation angle $\theta = \Omega\tau = 2\pi \times 250 \times \tau$, with τ in seconds.
Predicted $P(1) = \sin^2(\theta/2)$; $P(0) = 1 - P(1)$.
The 1 kHz CARRIER frequency and the assumed 250 Hz RABI rate describe different things: electrical oscillations versus state rotation.

Measured duration	Model rotation	Model P(1)	Model P(0)
1 ms	90° / π/2	50%	50%
2 ms	180° / π	100%	0%
3 ms	270° / 3π/2	50%	50%
4 ms	360° / 2π	0%	100%

Worked example: 2 cycles / 1,000 Hz = 0.002 s.
$\theta = 2\pi \times 250 \times 0.002 = \pi$. Therefore $P(1) = \sin^2(\pi/2) = 1$.
The model predicts an X rotation that flips $|0\rangle$ to $|1\rangle$ up to phase.

ENVELOPE AREA, NOT SIGNED SINE AREA

For resonant control, $\theta = \int\Omega(t)dt$. Complete sine cycles integrate to zero, so do not integrate the signed carrier voltage and call it the rotation. The assumed effective drive envelope is what matters.

10 / STUDENT WORKSHEET

Observe first. Calculate second.

Group: _____ Date: _____

Experiment	Setting	Measured result
A - amplitude	2 / 1 / 0.5 Vpp	Vpp: _____ / _____ / _____
B - frequency	0.5 / 1 / 2 kHz	Period: _____ / _____ / _____
C - pulse width	100 / 250 / 500 µs	Width: _____ / _____ / _____

D: cycles	Measured τ (ms)	Measured Vpp	Model P(1)
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____

1. What does the scope measure directly, and what is only modeled?

2. If f doubles with N fixed, what happens to burst duration N/f?

3. Why does changing phase look unchanged with one self-triggered trace?

11 / INSTRUCTOR GUIDE & BENCH FIXES

Keep the quantum link accurate.

This experiment prepares students to understand control signals; it does not operate a qubit.

Answers: A: measured height halves when Vpp halves. B: periods 2 ms, 1 ms, 0.5 ms. C: duty cycles 10%, 25%, 50%; levels stay 0/+1 V. D: durations 1-4 ms and predicted P(1) = 50%, 100%, 50%, 0% under the chosen calibration.

Discussion: voltage and timing are measured. Rabi rate, rotation and probabilities belong to the assumed model. Doubling carrier frequency at fixed N halves packet duration; a real qubit drive would also need to remain resonant. Phase appears fixed because self-triggering realigns each acquisition; observing relative phase needs another reference.

No trace? Check CH1 output ON, scope CH1 enabled, correct cable ports and Run / Single state.
Double voltage? Match generator High Z, scope 1 MΩ and 1:1 probe factor. Re-enter amplitude after changing the load.
No burst? Use triggered N Cycle, Burst ON and Manual source; press scope Single before generator Trigger.
Continuous signal? Burst may be OFF, or its source may be Immediate / Timer.
Missing packet end? Increase time/div or move trigger marker left.
Returning to A-C? Turn Burst OFF and return the scope to Run.

References and image credits
[1] Keysight, Trueform Series Operating and Service Guide: output load, burst count, burst phase and Manual triggering. 33500B / 33600A manufacturer manual.
[2] Keysight, InfiniiVision 3000 X-Series User Guide: inputs, triggers, Single, cursors and measurements.
[3] IBM Quantum Learning, Running Quantum Circuits: calibrated pulse frequency, strength, shape and duration.
[4] Quantum Machines, Rabi oscillations and gate calibration: control-pulse analogy. The particular 250 Hz model here is a classroom choice.
Official Keysight product photographs: 33500B datasheet and DSOX3014A product page. Screens in product photos are illustrative. Wave plots are calculated; real traces have finite edge times, noise and instrument tolerances. Menus vary by firmware. Predictions have not been physically validated on this bench.